MIL-R-52453C(ME)
31 May 1983
SUPERSEDING
MIL-R-52453B(ME)
14 April 1975

MILITARY SPECIFICATION

ROLLER, MOTORIZED, PNEUMATIC TIRED:

AIRMOBILE, 9 WHEEL

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pneumatic-tired, self-propelled, diesel engine driven roller, 9-wheel, capable of being airlifted by military helicopter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800	- Lubricating Oil, General Purpose, Preservative, (Water-Displacing, Low Temperature).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-R-3075	- Roller, Motorized, Road Diesel-or-Gasoline Driven, Packaging of.
MIL-A-8421	- Air Transportability Requirement, General Specification.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provision For Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 86 – Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

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(Application for copies should be addressed to the Superintendent for Documents, US Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF LABOR (DOL)

29 CFR 1926 - Occupational Safety and Health Administration (OSHA) Standards for Construction Jobs.

DEPARTMENT OF TRANSPORTATION (DOT)

FMVSS – No. 116 – Motor Vehicle Brake Fluids.

[The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.]

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z35.1 – Industrial Accident Prevention Signs, Specification For.

(Application for copies should be addressed to the American National Standards Institute, Inc. , 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1557 - Moisture-Density Relations of Soils using 10-lb (4.5 kg) Rammer and 18-in. (45.7 mm) Drop, Methods of Test For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC. (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, NW, Washington, DC 20005.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook

(Application for copies should be addressed to Tire and Rim Association, Inc., Command Building, 34 North Hawkins Avenue, Akron, OH 44313.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The roller shall be a wagon axle type, or articulated center hingepoint type. The roller shall be diesel engine driven, have nine variable pressure pneumatic tires, and be provided with full power hydraulic steering and oscillating wheels. The roller shall be of the latest model of the standard product of the manufacturer, and shall have been marketed commercially for at least one year prior to the opening of the technical proposal or bids. Rollers shall meet the requirements specified in table I.

Table I. Dimensions and Weights.

Height, Maximum	96 inches
Width, Maximum	80 inches
Empty Weight, Maximum	9,000 pounds

3.2 First article. Unless otherwise specified (see 4.3.1, 6.2) the contractor shall furnish one or more rollers for preproduction inspection within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce rollers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests, as specified (see 6.2). When specified (see 6.2), the contractor shall furnish one or more rollers for initial production inspection as specified in 4.3.2.

3.3 Material. Materials shall be as specified herein and on the drawings. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Material deterioration prevention and control. The materials used in fabrication of the roller shall be either inherently corrosion resistant and weatherproof or protected from the

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deterioration effect of exposure to weathering elements. The contractor and his sub-suppliers shall have material and material finishing methods utilizing the state-of-the art technology. These methods shall identify the specific materials, material finishes, and material treatments supplied on components and assemblies subject to weathering to provide protection against various forms of material deterioration in the roller's operating condition. These methods shall be part of, or an annex to the first article report. Any dissimilar metals used shall be in compliance with MIL-T-704 or MIL-STD-889.

3.4 Maintenance.

3.4.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using general purpose tools normally associated with equipment of this nature. After completion of any maintenance operation, the roller shall operate as specified herein.

3.4.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.5 Transportability. The roller shall be equipped with lifting attachments and shall be capable of being lifted and transported by helicopter without damage or permanent deformation. The roller shall be capable of being transported by truck trailers, rail, ocean-going carriers, C-130, C-141 and C-5A aircraft. Complete diagrams and instructions for lifting the roller and tying down the roller on freight cars and truck trailers shall be furnished in the operator's manual. Stencil or decal markings shall be applied to the roller at each lifting and tiedown point. The contractor shall provide sufficient data to the Government at the time of solicitation to confirm that the roller being supplied can be loaded and transported in the C-130, C-141 and C-5A aircraft. Cargo envelopes for transport in the required aircraft, are found in MIL-A-8421. Removal of any component to meet this requirement shall not be allowed.

3.6 Performance. The roller shall:

a. Be capable of compacting not less than 400 compacted cubic yards per hour of soil conforming to type GP or GW (Unified Soil Classification System) to at least 95 percent Modified AASHO in compacted lifts of not less than 6 inches.

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b. Be capable of negotiating a 10 percent grade over material offering a rolling resistance of 80 pounds per ton or greater for a total effective grade of 14 percent.
c. Be capable of operating in ambient temperatures of not less than 120° F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperatures (sump and gallery) and component manufacturer's hydraulic fluid temperature limits.

3.7 Human factors. The characteristics of the roller shall provide for operation by personnel ranging from the small man clothed, through the large man clothed in accordance with SAE J833.

3.8 Safety. All reciprocating, rotating, or moving parts of equipment shall be guarded when such parts may be inadvertently contacted by operating and maintenance personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. The roller shall meet the applicable requirements of the Occupational Safety and Health Administration.

3.9 Sound level. The sound level at the operator's station shall not exceed 85dB(A) when measured in accordance with SAE J919. The exterior sound level shall not exceed 88 dB(A) when measured in accordance with SAE J88 (see 4.3.3.9). Reference level is 20 µPa hereafter within this specification.

3.10 Body and frame. The body and frame shall be of one piece continuous construction or two separate bodies connected at a center hingepoint to provide articulated center hingepoint steering or an integral body and power unit with automotive steering. Rollers built in two sections with a hingepoint for articulation shall have a locking device to prevent articulation during lift and transport operations. A warning plate shall be posted on each side of the roller near the articulation point with the following words: "Articulation locking device shall be on when servicing or transporting this equipment. Caution: No Room for a person in this area when roller is turned." The body(s) shall be fabricated of steel plate and shall be reinforced with internal steel members attached to the body by continuous full length welds. The body shall withstand all forces encountered during performance at maximum load capacity without damage or permanent deformation. Accessible watertight cleanout openings shall be provided on the ballast body. Cleanout openings shall facilitate complete removal of the ballast and total drainage of water.

3.11 Engine. The roller shall be equipped with an industrial diesel engine that is furnished on commercial production rollers and which has been approved as complying with the

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Environmental Protection Agency (EPA) regulations governing "Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines". The engine shall be capable of meeting the performance requirement using diesel fuel conforming to VV-F-800. The operation of the roller as specified herein shall not require engine horsepower in excess of the engine manufacturer's continuous net rated SAE J8l6 horsepower of the engine furnished. The engine's governed speed shall be set and sealed in accordance with the engine manufacturer's recommendations.

3.11.1 Fuel tank. The fuel tank shall have a capacity sufficient to provide 8 hours operation at rated engine operating speed.

3.12 Power train.

3.12.1 Transmission. The transmission shall be of the power-shifted (hydrodynamic) continuous drive type within each speed range or hydrostatic type.

3.12.1.1 Power-shift transmission. A torque converter shall be supplied with the powershift transmission. The transmission and torque converter fluid shall be filtered by full flow filter(s) having replaceable elements.

3.12.1.2 Hydrostatic transmission. A filter element shall be provided to filter all the oil and shall be installed in the suction line between the reservoir and the transmission. The filter element shall not be used at a flow rate in excess of that at which it was tested.

3.12.2 Differential. The roller shall be equipped with a no-spin or bevel gear differential, with roller chain or propeller shaft drive system or both.

3.13 Suspension system. The roller shall be equipped with either vertical axle oscillation or tilt axle oscillation. Each wheel or wheel pair of at least one axle shall oscillate independent from the other wheel or wheel pair. On axles, or bolsters having five wheels, the fifth wheel shall be capable of independent tilt or vertical oscillation.

3.14 Hydraulic system. Hydraulic power-assist steering or hydraulic steering for heavy-duty vehicles shall be installed and shall include the hydraulic fluid reservoir, hydraulic cylinder(s), pump, filter, pressure relief valve, and all necessary controls, lines, and fittings to perform steering operations.

3.15 Brakes. The roller shall be equipped with service brakes, emergency brakes, and parking brakes. The brake system shall be in accordance with SAE J1136. Brakes which normally use glycol brake fluid as medium shall be serviced with DOT FMVSS No. 116, DOT 5 type fluid meeting the minimum requirements of MIL-B-46176 for Di-organopoly-siloxane fluid (silicones). DOT 5 type fluids shall include all non-petroleum based fluids for straight hydraulic or air hydraulic brakes, and operator's controls.

3.16 Tires and rims. Tires shall be of the code C-1, smooth tread design for compaction service. Size of tires shall be 7.50 x 15, with a ply rating of not less than 12. Tires shall have

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individual rated load-carrying capabilities based on maximum individual wheel loading and related inflation pressure required to obtain maximum compactive force. Tires, tubes, flaps, and wheels shall conform to Tire and Rim Yearbook.

3.16.1 Tire pressure regulating system. The roller shall be equipped with a central tire inflation system controlled from the operator's seat. The system shall be capable of varying the pressure in each tire from 40 psi to 120 while at 4 standstill or traveling.

3.16.2 Tire wetting system. A system will be provided for wetting and cleaning tires. The wetting system shall be the pressurized type and shall include 8 water pump, tank of at least 75 gallons, cocoa mats or scrapers, and all necessary controls to wet all tires. The system shall be operable from the operator's platform.

3.17 Electrical system.

3.17.1 Lighting system. A complete 12-volt lighting system shall he provided. The lighting system shall include two forward and two rearward directed sealed-beam floodlights and two red lights at the rear end. Independent controls shall he provided for forward directed floodlights, for rearward directed floodlights, and for rear-mounted red lights. The light controls shall be clearly identified.

3.17.2 Floodlights. The floodlights shall conform to SAE J598, PAR 46, and shall be provided with lamp housings, except for mounting brackets. Mounting brackets shall be made of steel not less than 1/4 inch thick and shall provide an adjustment of not less than 30 degrees above and 30 degrees below the horizontal in the vertical plane and not less than 180 degrees to the right and 180 degrees to the left in the horizontal plane.

3.17.3 Red lights. The red lights shall be 4-inch nominal diameter, shall be waterproof, and shall conform to SAE J759, S-AI-T for 12-volt use. The stoplight filament of each red light shall be connected so as to be actuated when the service brakes are applied. The red lights shall be mounted as far apart as possible at the rear of the ballast body.

3.17.4 Wiring. Cable for wiring shall conform to SAE J82l. Grounded return electrical circuits shall be used and shall be tagged at the connections with band markers. Grommets shall be installed where wiring passes through structural steel sections and sheet metal to prevent chafing of the cable.

3.18 Operator's compartment.

3.18.1 Seat. The contractor's standard waterproof commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls, and visibility of the work zone and all instruments and gages mounted in the operator's compartment when seated.

3.18.2 Controls. The contractor's standard commercial instruments and controls shall be provided. Related instruments and controls when optional systems are required shall be

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provided. Instruments and controls shall be located in the operator's compartment and shall be within reach of the operator in his normal operating position.

3.18.3 Ladders or steps. Ladders or steps, and handholds shall be provided in such quantity and of such size that entrance thereto and exit therefrom may be unhampered and non-hazardous. Handholds will also be conveniently located to provide stability at the operator's compartment.

3.19 Lifting and tiedown attachments.

3.19.1 Lifting attachments. Permanently affixed attachments that enable the roller to be lifted in its normal travel position shall be provided. The attachments shall meet the requirements specified in MIL-STD-209. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.19.2.

3.19.2 Tiedown attachments. Permanently affixed attachments that permit fastening the roller to the floor or deck of a transportation medium shall be provided. The attachments shall meet the requirements specified in MIL-STD-209.

3.20 Electromagnetic Interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the roller shall conform to SAE J55l.

3.21 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173 or a varnish having 1 percent of copper-8-quinolinolate or 6 percent galicylanilide (by weight) based on the nonvolatile content of the varnish, with the following exceptions:

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component and element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.22 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. A lubrication chart in accordance with SAE J753 shall be furnished with each roller.

3.22.1 Engine lubrication. The engine lubrication shall be in accordance with MIL-L-2104.

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3.22.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun having a minimum capacity of 15 ounces and equipped with a straight hydraulic coupler and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.22.3 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.22.4 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipstick, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain outlet shall be fitted with a drain plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure.

3.22.5 Grease lubrication. All grease lubrication shall be with greases conforming to MIL-G-10924, MIL-G-23827, or SAE J754. The roller shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.22.6 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with winter grades of military oil designated for use in the temperature range of 0° F to 32° F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.22.7 Servicing and adjustments. The contractor shall service and adjust each roller for correct operation and travel including at least the following: Adjustment of the engine and of the roller machinery; adjustment of the brake system and electrical system; inflation of tires; and complete lubrication of the roller with the types and grades of lubricant recommended.

3.23 Oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

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3.24 Treatment and painting. The parts of the roller normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. Unless otherwise specified (see 6.2), the final coat shall be in accordance with MIL-C-46168.

3.25 Data plates. Data plates shall be applied to the equipment on metal plates in accordance with the "Metal Name Plate Industry Standards and Practices", as published by the National Association of Metal Name Plate Manufacturers, Inc., (NAME). Identification and shipping data plates shall be of material in accordance with MIL-P-514, Composition C. The plates shall be secured to each end item in accordance with the NAME Standards and Practices.

3.25.1 Identification. The equipment shall be identified in accordance with MIL-STD-130 and MIL-P-514. The date of the delivery shall be that of the complete equipment under contract and shall not be prior to date of Government acceptance.

3.25.2 Instruction plates. The equipment shall be equipped with instruction plates, decals, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing. The safety, warning, caution, and notice plates shall be in accordance with ANSI Z35.1 and compatible with the space available.

3.25.3 Lifting and tiedown instruction plates. Instructions for the use of the attachments or procedures shall be included on a metal plate permanently affixed to the equipment. The plate shall include a silhouette of the item together with the center of gravity.

3.25.4 Shipping data plates. Shipping data plates for each roller shall be furnished and shall show the silhouette of the roller indicating the location and capacity of tiedown provisions, slinging provision, extraction provisions, suspension provisions, the center of balance and shipping weight. Plates shall conform to MIL-P-514, Type III or contractor's standard plates subject to the approval of the contracting officer. The plates shall be attached to the left side of the roller and shall show the locations as viewed from the left side. The plate shall be located in conspicuous protected locations. All plates shall be permanently attached.

3.26 Marking. Marking and registration shall be in accordance with MIL-STD-642. Color shall be lusterless black.

3.27 Toolbox. A lockable storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided.

3.28 Workmanship.

3.28.1 Steel fabrication. Steel used in the fabrication of the roller shall be free from kinks and sharp bends. Steel plate shall not be warped. The straightening of material shall be done by methods that will not cause injury to the metal. Burned surfaces of flame-cut materials shall be ground or machined to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

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3.28.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washer or lockwashers shall be provided and all bolts, nuts, and screws shall be tight.

3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes and in full contact with the surface of the member.

3.28.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof of service loading.

3.28.5 Surfaces. All parts, components, and assemblies including castings, forgings, molded parts, stampings, bearing, seals, machined surfaces, and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Preproduction roller inspection. Unless otherwise specified in the contract (see 6.2), the preproduction roller shall be examined as specified in 4.4 and tested as specified in 4.3.3.1 through 4.3.3.9 to determine conformance to the requirements of this specification. Presence of one or more defects shall be cause for rejection. Failure of any test shall be cause for rejection.

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4.3.2 Initial production roller inspection. When specified (see 6.2), one or more rollers will be selected at random by the Government from the rollers being produced and will be examined as specified in 4.4 and tested as specified in 4.3.3.1 through 4.3.3.9 to determine conformance to the requirements of this specification. Acceptance of an initial production roller shall not exclude the remaining roller from the quality conformance inspection and acceptance provisions specified in Section 4.

4.3.3 Tests.

4.3.3.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The roller shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.3.3.2 Compaction rate. The fill material shall correspond to Group GP or GW of the Unified Soil Classification System conforming to SAE J106. The compacted fill strip shall be rectangular, and not less than 24 feet wide. The fill may be placed prior to the start of compaction operation or during compactions. Before testing is started, soil's moisture content is to be determined and a test strip of the material is to be rolled. Hauling equipment shall not be run over spreaded uncompacted fill material before or after dumping the material. Spreading of fill material shall be done with a grader. Spreading runs shall be only that necessary to reach the estimated uncompacted fill lift. Water distribution equipment may be used as required. The material to be compacted shall be of sufficient thickness to give a compacted lift of at least 6 inches. The laboratory density of the soil to be compacted shall be determined in accordance with ASTM D1557, method optional. The Government reserves the right to witness any laboratory testing. When the contractor feels that compaction of 400 compacted cubic yards of fill to a density of 95 percent Modified AASHO has been accomplished, he may stop compaction operations. Time used in density testing shall not be counted as compaction time. Two density samples shall be taken at each 25 foot station. The first sample shall be taken 7 feet left of the centerline, and the second sample 9 feet right of the centerline. This pattern shall be repeated for the length of the fill. In addition, another 3 samples shall be taken at random locations every 30 feet. Volume of compacted soil shall be determined as follows:

a. Stake out the fill in 25 foot stations.
b. With a surveyor's level and rod, determine the elevations of the left edge, right edge, and centerline at each station before placing the fill and again after compaction. Determine the compacted lift thickness.
c. Determine the compacted width at each station.
d. Compute the volume using the average end area method.

A compaction rate of less than 270 compacted cubic yards per hour and failure to achieve a compacted lift thickness of at least 6 inches shall constitute failure of this test. Density testing

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shall be accomplished by a nuclear density meter. The nuclear density tester shall have been calibrated against the soil to be compacted, if applicable. The Government shall witness such calibration. An average density of less than 95 percent Modified AASHO shall constitute failure of this test. A compaction rate of less than 270 compacted cubic yards per hour and failure to achieve a compacted lift thickness of at least 6 inches shall constitute failure of this test.

4.3.3.3 Gradeability. Operate the roller on a grade of not less than 10 percent over fill material having a rolling resistance of not less than 80 pounds per ton (total effective grade of 14 percent) or, operate the roller up a grade of at least 14 percent with a smooth surface. Inability to negotiate the grade under either condition specified above shall constitute failure of this test.

4.3.3.4 Brake system test. The service brake system and emergency stopping system shall be tested in accordance with SAE J1136. Inability to perform as specified shall constitute failure of this test.

4.3.3.5 High temperature operation. The contractor shall furnish test data to prove that the roller is capable of operating in an ambient temperature of 120ºF without exceeding the engine manufacturers recommended limits for radiator top tank temperature, engine sump and gallery oil temperatures and hydraulic component manufacturers hydraulic fluid temperature limits.

4.3.3.6 Test procedures of lifting provisions. Test provisions shall be as follows:

a. Lift the roller and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the roller. Determine the forward angles of application for each provision. Measure the clearance between each sling and the roller. Measure eye openings and clearance dimensions of the provisions. Measure the load carried by each provision.
b. Restrain the roller by anchoring the main frame and subject each provision to a force equal to the load factor specified in MIL-STD-209 times the load it carried in paragraph (a) above. An alternate method of test is to lift the roller as in (a) above and add weights to the main frame until the required force is obtained. Hold the roller under either of the above loading conditions for at least 90 seconds.

4.3.3.6.1 Failure criteria. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209, shall constitute failure of this test.

4.3.3.7 Test procedures of tiedown provisions. With the roller anchored by means other than the provisions being tested, apply the proportion of the static loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown. Failure to meet this requirement shall constitute failure of this test.

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4.3.3.8 Electromagnetic interference. Unless otherwise specified, the roller shall be tested in accordance with SAE J55l. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the preproduction model test report. Failure to meet this requirement shall constitute failure of this test.

4.3.3.9 Sound level measurement. Unless otherwise specified the sound level shall be conducted (see 6.2 and 6.7). Frequency measurements, dB(A) levels, instrument identification, environmental information, and main sound source identification shall be reported.

4.3.3.9.1 Failure criteria. Failure to meet the criteria specified in 3.9 shall constitute failure of the test.

4.4 Quality conformance inspection.

4.4.1 Examination. Each roller shall be examined for the characteristics as listed in Table II. Presence of one or more defects shall be the cause for rejection.

Table II. Examination schedule.

Number	Characteristic	Requirement Paragraph
101.	Engine not as specified.	3.11
102.	Power Train system not as specified.	3.12
103.	Suspension system not as specified.	3.13
104.	Hydraulic system not as specified.	3.14
105.	Brake system not as specified.	3.15
106.	Tires and rim not as specified	3.16
107.	Body and frame not as specified	3.10
108.	Operator's compartment not as specified.	3.18
109.	Fuel tank not as specified.	3.11.1
110.	Electrical system not as specified.	3.17
111.	Fungus and moisture resistance not as specified.	3.21
112.	Lifting and tiedown attachments not as specified.	3.19
113.	Oil sampling valve not as specified.	3.23
114.	Safety not as specified.	3.8
115.	Lubrication not as specified.	3.22
116.	Transportability not as specified.	3.5
117.	Ease of maintenance not as specified.	3.4
118.	Lighting not as specified.	3.17.1
119.	Materials not as specified.	3.3
120.	Toolbox not as specified.	3.27
121.	Data plates not as specified.	3.25
122.	Treatment and painting not as specified.	3.24
123.	Marking not as specified.	3.26
124.	Workmanship not as specified.	3.28

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Table II. Examination schedule. (Continued)

Number	Characteristic	Requirement Paragraph
125.	Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.	3.3.1
126.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1
127.	Contractor does not have documentation available for identification of material, material finishes or treatments.	3.3.1

4.4.2 Operational run-in. Each item of equipment shall be serviced as required and shall be run-in in accordance with the manufacturer's normal procedures for a period of not less than 60 minutes. One or more functional defects shall constitute failure and acceptance of the equipment shall not be approved until all defects are corrected.

4.5 Inspection comparison. When initial production inspection is specified the Government may select rollers at any time during the contract production period and subject these rollers to the examination specified in 4.4.1 and the tests specified in 4.3.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction roller and initial production roller. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5.1 Inspection failure. Failure of an inspection comparison roller to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3.3 shall be cause for refusal by the Government to continue acceptance of production rollers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on rollers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all rollers accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined in accordance with the quality assurance provisions of MIL-R-3075.

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5. PACKAGING

5.1 Preservation, packing and marking. Each complete roller shall be preserved, packed, and marked in accordance with MIL-R-3075. The level of preservation shall be level A, B, or commercial and the level of packing shall be level A, B, or commercial as specified (see 6.2).

6. NOTES.

6.1 Intended use. The roller is intended for use in compacting granular soils.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of specification.
b. Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
c. When first article inspection is not required (see 3.2).
d. Time frame for submission of first article roller and number of rollers required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When preproduction roller inspection is not required (see 3.2).
g. When initial production roller inspection is required and number of rollers to be furnished (see 3.2).
h. When sound measurement tests are not required (see 3.9).
i. When electromagnetic interference requirements are other than as specified in 3.20.
j. When fungus and moisture resistance is not required (see 3.21).
k. When the finishing coat shall be other than as specified in 3.24.
l. Degree of preservation and degree of packing required (see 5.1).

6.3 First article. Any changes or deviation of production rollers from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish rollers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publication, instructional materials, illustrated parts lists, and contractors maintenance and operation manual to be furnished with each roller.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of rollers.

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6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.7 Sound level. When the sound level of the equipment exceeds 85 dB(A) at the operator's station, a noise hazard caution sign per MIL-STD-1474 shall be permanently affixed to the equipment in the operator's compartment. The 85 dB(A) contour should be defined. The noise hazard zone shall be clearly stenciled on the equipment in a conspicuous location with 3/4 inch letters stating: "Hearing protection required within X feet from this equipment".

Custodians:
Army - ME

Preparing activity:
Army - ME

User activity:
Army - AT

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